3.2.S.3.2. Impurities

The solid phase phosphoramidite chemistry used to manufacture 2439-exNA drug substance means that all low molecular weight organic impurities such as impurities originating from the raw materials that do not react with the with the growing oligonucleotide will be purged by the process during the wash steps after the coupling steps. Additional purging occurs during the purification of the strands and during the UFDF steps. Impurities that could be present in the 2439-exNA Drug Substance include oligonucleotide related impurities, residual solvents, and elemental impurities. The batch used for toxicological study and formulated to drug product for proposed clinical use are identical meaning that all impurities present are identical and therefore qualified.

3.2.S.3.2.1. Oligonucleotide Related Impurities

Potential oligonucleotide related impurities arising from the route of synthesis of the 2439-exNA antisense and sense strands are comparable those found in all nucleic acid oligonucleotides and include:

- Deletion “failure” sequences: [N-1], [N-2], [N-3], etc.
- Addition sequences: [N+1 or more]
- Impurities with a modification to a sugar residue, base residue, internucleotide linkage, and residual/incomplete removal of protecting groups
- Sequences with PO linkage instead of phosphorothioate (PO impurities)

Oligonucleotide impurities originating from the solid phase oligonucleotide syntheses are efficiently reduced during the ion exchange purification chromatography step of the oligonucleotide strands manufacturing process by using an appropriate fraction pooling strategy that ensures the 2439-exNA single strand specification is met. The pooling strategy is based on and confirmed by in-process controls of the chromatographic fractions.

All impurities present in the final 2439-exNA drug substance would fall into one of the four possible classes of impurities described above and are controlled as part of the 2439-exNA Drug Substance release specification.

According to ICH Q3A (R2), limits on the content and qualification of impurities in new drug substances produced by chemical syntheses should be progressively set while the development of the drug progresses towards registration applications. ICH Q3A (R2) does not cover oligonucleotides however, we propose to classify impurities in 2439-exNA as detailed in the recommendations by Capaldi et al. 2017 during the development lifecycle of the IND as follows:

- Major impurities, with a level greater than 1.5% (qualification).
  - Molecular weight determined by liquid chromatography-mass spectrometry (LC-MS)
  - Structure proposed, based on molecular weight.
  - Synthetic pathway responsible for generating impurity will be proposed, if possible.

  - Relative retention time will be established by the ultra-performance liquid chromatography (UPLC) method used for release testing.
  - Qualified initially based on pre-IND GLP tox studies.

- Major impurities, with a level greater than 1.0% and up to 1.5% (identification).
  - Molecular weight determined by liquid chromatography-mass spectrometry (LC-MS)
  - Structure proposed, based on molecular weight.
  - Synthetic pathway responsible for generating impurity will be proposed, if possible.
  - Relative retention time will be established by the ultra-performance liquid chromatography (UPLC) method used for release testing.
- Minor impurities, defined as greater than 0.2% (reporting threshold based on UPLC limit of quantitation [LOQ]), and up to 1.0%.
  - Relative retention time will be established using HPLC method used for release testing.

Oligonucleotide related impurities will receive limits based on data obtained from analysis of 2439-exNA Drug Substance batches made by a process that closely resemble the intended commercial processes in later stages of the development.

The identification of the chemical nature of peaks corresponding to Major impurities will be assessed in real time as additional batches are manufactured. Molecular mass of the most abundant impurities will be measured to determine their most likely structure.

All reported impurities are identified by their relative retention time (RRT) in the HPLC analysis. Table 1 and 2 compare the quantity of impurities above XXX of the 2439-exNAA and 2439-exNAS peak in the UPLC analysis and summarizes proposed structures based on analysis of LC-MS data from clinical lot XXX
Representative chromatograms for the 2 single strand batches are also provided in Figure 3 and Figure 4.

3.2.S.3.2.2. Residual Solvents

Organic solvents are used during the Synthesis and Deprotection Steps (Section 3.2.S.2.2). The main solvent used is XXX. This solvent and the others are class 2 according to ICH Q3C(R6). Based on existing results the solvents are efficiently purged during the wash steps in synthesis, the purification, the ultrafiltration and the lyophilization for the volatile ones. The solvents are adequately removed during the final freeze drying that is performed under extremely low pressure, a process that removes any residual volatile solvents to low levels.

The acceptance criteria for the tested solvents are set according to the ICH Q3C Guideline and residual solvent levels are below the acceptance criteria (Section 3.2.S.4.1).

Table 7: Residual Solvent Test Evaluation

Solvent	ICH Limit (ppm)	Result (ppm)

3.2.S.3.2.3. Elemental Impurities

The synthesis process for 2439-exNA Drug Substance does not include any metal catalysts, and therefore, contamination with elemental impurities is unlikely. The cGMP batch has been tested by inductively coupled plasma-mass spectrometry (ICP-MS) for the presence of elements possibly originating from the primary material of construction for the equipment used (stainless steel will contain: V, Cr, Fe, Co, Ni) as well as the four Class 1 elements (As, Cd, Hg, and Pb). The batch analysis has revealed elemental impurity levels below the ICH Q3D option 1 limits (parenteral) for each element (please refer to –Section 3.2.S.4.4).

Table 8: Elemental Impurity Detection Evaluation

Element	Classification	ICH Limit (ppm)	Result (ppm)
V	2A	1	
Cr	3	110	
Fe	N/A	N/A	
Co	2A	0.5	
Ni	2A	2	

As	1	1.5	
Cd	1	0.2	
Hg	1	0.3	
Pb	1	0.5	

3.2.S.3.2.4 Genotoxic Risk Assessment

The genotoxicity of reagents and solvents used during the manufacture of 2439-exNA drug substance were evaluated. 2439-exNA is an oligonucleotide duplex with significant differences in MW, structure and chemical properties from that of small molecule reagents and solvents. The differences allow efficient removal of the reagents and solvents during purification steps. The risk of genotoxicity in 2439-exNA drug substance from reagents and solvents is low. An AMES study was conducted on 2439-exNA and summarized in Section 2.6.6.

All raw materials, intermediates, and expected potential impurities in the manufacturing process of 2439-exNA drug substance were assessed for their mutagenic potential.

3.2.S.3.2.5 Nitrosamine Risk Assessment

At this stage of the development the Sponsor has not conducted a formal risk assessment of the potential of nitrosamines forming during the drug substance and drug product manufacture of 2439-exNA. The potential risk of nitrosamine impurities can be discussed in the context of the potential risk of introduction and formation of nitrosamines during the manufacture of the

2439-exNA, the patient exposure to these potential impurities and any supporting non-clinical study.
2439-exNA Drug Substance is manufactured with a platform-based process typically used to produce therapeutic oligonucleotides. A platform solid-supported chemical synthesis followed by cleavage and deprotection, followed by a platform purification process using chromatographic separations, Tangential Flow Filtration (TFF), concentration, and lyophilization is used for the manufacture of the duplex drug substance. The drug product manufacturing process for the manufacture of 2439-exNA drug product uses a standardized process and excipient used for the fill-finish manufacture processing.

Due to the platform nature of synthesis, oligonucleotides share a range of common impurity types, including both product and process related impurities. Incorporation of potential impurities, such as nitrosamines, may be introduced by starting materials and reagents and/or possibly formed during the manufacture of intermediates. A comprehensive discussion regarding the potential risk conditions for the formation of nitrosamines in the end-to-end manufacture of oligonucleotides using platform-based processes for the drug substance and drug product manufacture applicable to the 2439-exNA manufacture is summarized by Borths 2022 in an industry consortium authored white paper regarding the nitrosamine risk assessments in oligonucleotides. The risk of introducing and/or forming nitrosamine impurities is low in terms of raw materials, synthesis and purification and the overall nitrosamine risk in terms of the overall end-to-end manufacturing process is low.

3.2.S.3.2.6 REFERENCES

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